

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME:Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I _Y.Siva Pardhav_(192412612)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that: Constraints:

- i. D1 cannot work Night shift ii. D2 must work before D3 (Morning < Afternoon < Night) iii. Only one doctor per shift iv. D3 cannot work Morning v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1 iii. Cannot pass through obstacles iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

- d) Justify why your chosen approach is the most suitable for solving this problem under the
- Clearly identify all constraints

Question 1

Problem Understanding

1. 3 doctors, 3 shifts.
2. All constraints identified.

Constraint Analysis

1. $D1 \neq \text{Night}$.
2. D2 before D3.
3. $D3 \neq \text{Morning}$.
4. One doctor per shift.

Solution Development

Step	Assignment	Result
1	D1 = Morning	Valid
2	D2 = Night	Invalid
3	Backtrack	Remove D2 = Night
4	D2 = Afternoon	Valid
5	D3 = Night	Valid

Final Schedule

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

Answer: D1 = Morning, D2 = Afternoon, D3 = Night.

Question 2

Constraints

1. Move Up, Down, Left, Right
2. Cost = 1 per move
3. Cannot cross obstacles
4. Manhattan Distance heuristic

BFS Search Process

1. Start at S
2. Expand neighboring nodes
3. Add valid nodes to queue
4. Ignore obstacles
5. Continue until Goal G is reached

Example

Queue = [S]

Expand S

Queue = [A, B]

Expand A

Queue = [B, C]

Expand B

Queue = [C, D]

Expand C

Queue = [D, G]

Expand G

Goal Reached

Manhattan Distance

$$h(n) = |x_1 - x_2| + |y_1 - y_2|$$

Optimal Path



Cost

$$3 \times 1 = 3$$

Answer: BFS finds the shortest path with minimum cost.

Question 3

(a) Constraint Analysis

Obstacles

1. Robot cannot move through blocked cells.

Limited Sensing

1. Robot can only observe adjacent cells.
2. Environment is partially observable.

Risky Zones

1. Normal cost = 1
2. Risky zone cost = 3

Dynamic Knowledge

1. Robot updates map while moving.

(b) Solution Development

Chosen Strategy

Online Uniform Cost Search (UCS)

Steps

1. Start at S
2. Sense neighboring cells
3. Update internal map
4. Calculate path costs
5. Choose lowest-cost path
6. Move
7. Replan if obstacle discovered
8. Repeat until G

(c) AI Concepts

Intelligent Agent

1. Sensors gather information.
2. Actuators perform movement.
3. Makes decisions autonomously.

Rationality

The robot chooses the safest and lowest-cost path.

Environment Type

Property	Type
Observability	Partially Observable
Agent	Single Agent
Actions	Sequential
Environment	Dynamic
States	Discrete

(d) Justification

Uniform Cost Search is most suitable because:

1. Handles different path costs.
2. Finds minimum-cost solution.
3. Avoids risky areas when possible.
4. Works well with online replanning.
5. Suitable for partially observable environments.

Conclusion

The rescue robot acts as a rational intelligent agent and uses Online Uniform Cost Search to safely reach the survivor while minimizing total cost.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided